Muhammad Haroon Ali

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Profile

Computer Science student who builds things to understand them — from a compiler with its own RISC-V backend to applied ML pipelines. Fast learner, comfortable with ambiguity, and looking to bring that same hands-on curiosity to a software engineering or AI/ML Werkstudent team.

Education

Saarland University — Saarbrücken, Germany
B.Sc. in Computer Science — *Oct. 2024 – Present*

- Relevant coursework: Data Structures & Algorithms, Software Engineering, Object-Oriented Programming, Machine Learning & Data Analysis, System Architecture, Statistics, Linear Algebra, Functional Programming

Experience

Teaching Assistant — Statistics for Data Science & AI — Saarbrücken, Germany
Saarland University — *Apr. 2026 – Sep. 2026*

- Led weekly tutorial sessions and designed course materials and assessments for a Statistics for Data Science & AI course
- Evaluated the response quality of an in-house LLM chatbot (ALA project) by systematically auditing outputs against course material and expected reasoning
- Designed structured test prompts to surface hallucinations and inconsistent reasoning, logging and categorizing recurring failure patterns for the development team
- Fed evaluation findings back into the model's iteration cycle, improving alignment between chatbot responses and course content

Projects

TinyC Compiler & Verification System | *Java, RISC-V Assembly, Static Analysis, JUnit*

- Architected a multi-pass compiler in Java for a subset of C – pointers, static scoping, global declarations, and control flow – with a custom AST and multi-phase static analysis (name resolution, scope management, type checking, source-location diagnostics)
- Engineered a RISC-V backend translating semantically validated ASTs into executable assembly, handling stack frame management, register allocation, pointer dereferencing, and function prologue/epilogue sequences
- Built an automated formal verification layer that extracts logical Verification Conditions (VCs) from annotated code to prove total and partial program correctness

RayCast | *C, Computer Graphics, Systems Programming*

- Built a 2.5D rendering engine from scratch in pure C with zero external libraries, using raycasting to render 2D grid maps as interactive first-person perspectives
- Modeled a pinhole camera for FOV and ray-wall intersection calculations, with custom texture mapping via hand-written .ppm image processing and a streaming stdin/stdout frame pipeline

Ultimate Tic-Tac-Toe Engine | *Java, Object-Oriented Programming, Software Architecture*

- Architected a 2-tier nested board system using custom matrix data structures to manage sub-board states and global board transitions
- Designed a real-time move validation and constraint engine that dynamically routes turns based on nested gameplay rules
- Applied OOP principles (encapsulation, polymorphism, abstraction) and the MVC pattern to cleanly separate game logic from UI and event handling

Titanic Survival Prediction | *Python, scikit-learn, pandas, NumPy*

- Engineered a Random Forest classification model to predict passenger survival, achieving ˜79.89% accuracy
- Performed feature engineering, missing-data imputation, and hyperparameter tuning; evaluated performance using standard classification metrics

Technical Skills

Languages: Java, Python, C, C++, JavaScript, SQL, HTML/CSS, OCaml, RISC-V Assembly
AI/ML & Data: scikit-learn, pandas, NumPy, matplotlib, Machine Learning, Statistical Analysis
Software Engineering & Tools: Git/GitHub, Linux, Object-Oriented Design (MVC), Unit Testing (JUnit), Software Architecture